ArtsibEro:

An Intuitive and Accessible Document Management System for a Modern School System

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I. Customer Statement of Requirements

Project Motivation and Background

At General Roxas Elementary School (GRES), the admin and faculty work hard to provide quality education and service for their students. However, the school heavily relies on manual, paper-heavy processing for managing student records and fulfilling admin tasks. This practice is often tedious and tiring for both the school system's faculty, staff and the students' guardians.

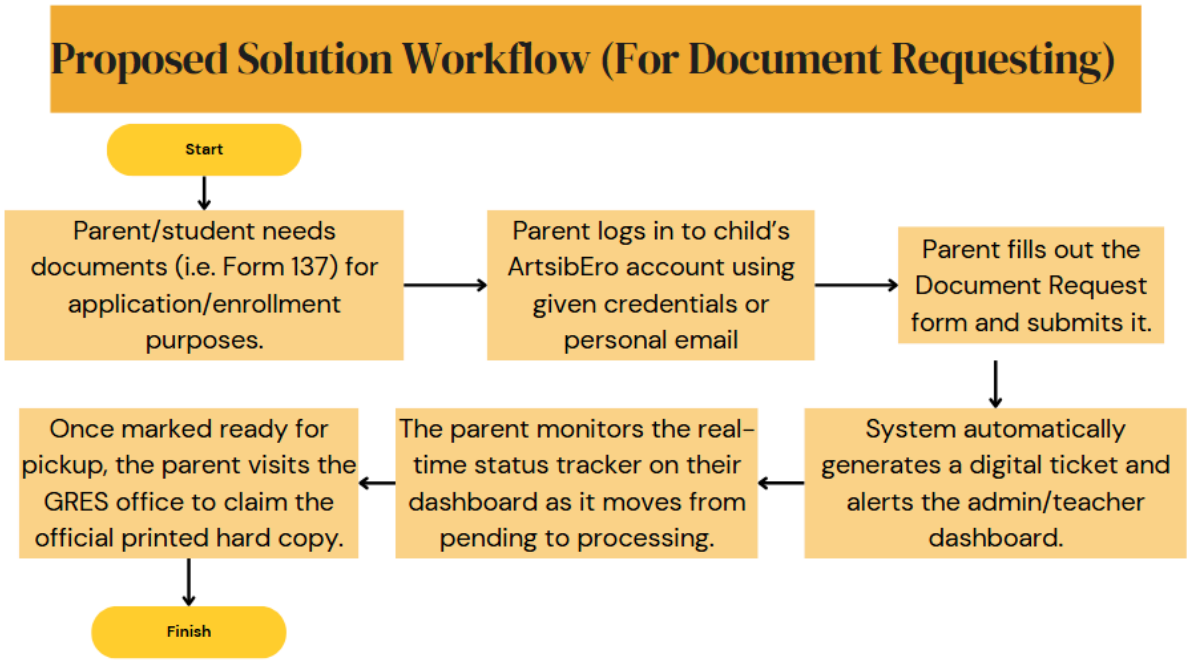
For parents and guardians of GRES, securing a simple school document for transfer or applications (such as Form 137, Certificate of Good Moral Character, or Certificate of Enrollment) often requires unnecessary travel, wait times, and strict office hours just to get paperwork verified, signed, and released. However, because formal admin channels are complicated, requests often land on the classroom teachers, which causes them to manually locate, photocopy, or send pictures of hardcopy records to the parents. Teachers are also often forced to encode the same student information, attendance logs, and raw scores across multiple platforms. Instead of having an automated system, teachers spend time constantly fixing custom spreadsheets or templates and copying data from previous formats just to get their grade computation done. Recent educational research emphasizes that public school teachers face challenges because of administrative workloads and excessive paperwork that drain their time for actual teaching and increase burnout (Flores & Gumban, 2026; Inquirer). When the registrar or admin needs to collect records for a Form 137, they must manually sort and find through physical cabinets and re-verify that teachers have already recorded elsewhere in previous semesters. All of this points to an inefficient system that is in need of a fast and reliable solution.

Currently, the Department of Education (DepEd) uses the Learning Information System (LIS) as a national database for student enrollment and macro-level tracking for transferring learner records (DepEd Order No. 54, s. 2016). However, while this system exists, studies assessing the LIS reveal that local schools still experience inefficient workflows (Bariquit, 2024). The LIS is strictly an administrative tool, meaning parents cannot log into it to request their child's documents, and teachers cannot use it as a grading sheet. Since there is no localized

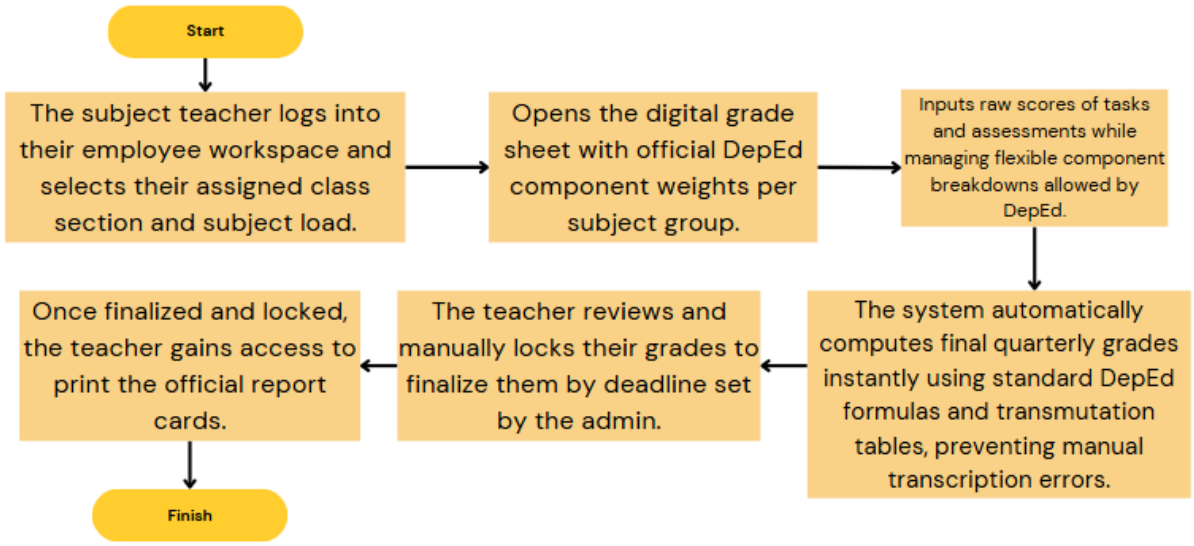
system connecting the parents directly to GRES's office, the process of requesting and formatting documents is still mostly manual.

The Proposed Solution

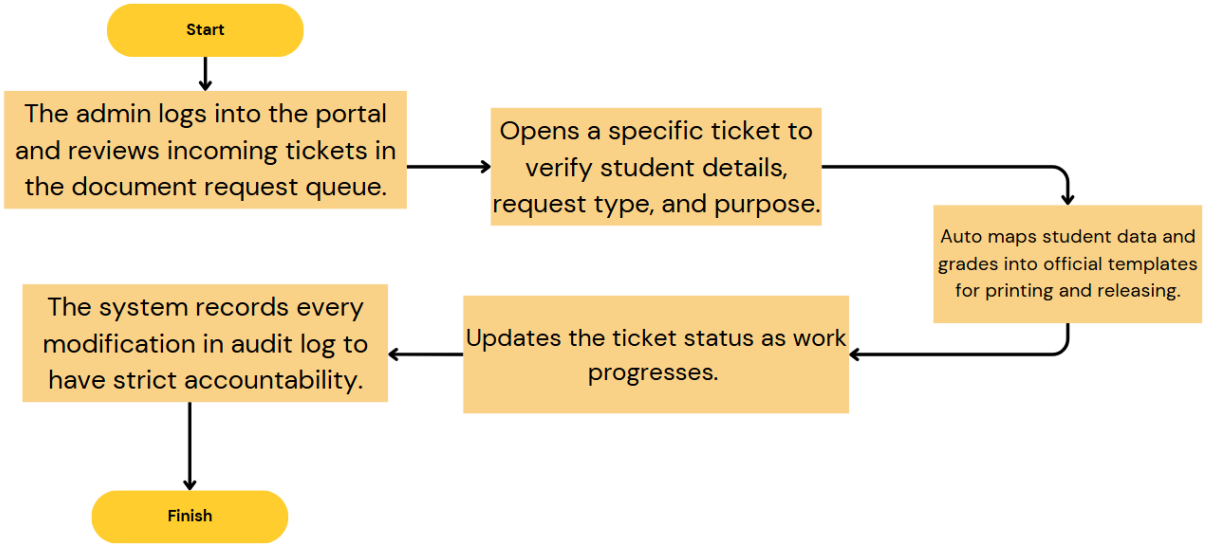
To address this, our group proposes a **centralized online dashboard** called **ArtsibEro** that simplifies and streamlines academic record-keeping, daily attendance tracking, document management and document-request processing. Ultimately, the system could replace the slow and manual, paper-heavy workflows with a digital space that automates grade computations, manages communication lines, and handles document tracking. This would reduce the time that teachers and admins spend sorting through physical files, and spare the guardians the hassle of lining up at the school's offices. Our proposed website aims to, first and foremost, digitalize school documentation for ease of requesting and archiving. We will consider existing documents and processes to facilitate an efficient transition.



Proposed Solution Workflow (Grading)



Proposed Solution Workflow (Admin Document Processing)



System Requirements

1. Parent and Guardian Features

- Allow parents/guardians to log into the website for the first time using initial/temporary credentials such as the child's Learner Reference Number (LRN) and a password (with strict data privacy controls).
- Allow parents/guardians to link their personal email accounts to their profile after initial login, enabling them to sign in directly with their email for easier access.
- Display a digital bulletin board on the dashboard for school news, enrollment schedules, and application opportunities.
- Allow parents/guardians to view their child's past grades and academic records.
- Allow parents/guardians to track their child's attendance records.
- Allow parents/guardians to submit an online request for specific school documents (such as Form 137).
- Display a visual status tracker so parents/guardians can see if their document is pending, processing, or ready for pickup.
- Allow parents/guardians to send direct inquiries or messages to their child's subject teachers or class advisers regarding academic performance, attendance, or document requests

2. Teacher and Grading Features

- Allow teachers and admin to log into a workspace using their employee accounts.
- Provide a digital attendance log for advisers to easily record and update daily student attendance.
- Provide teachers with a digital grade sheet to encode and update student scores, featuring predefined DepEd component weights per subject group while allowing flexibility in managing itemized tasks and assessments within those components.
- Automatically calculate final quarterly grades as soon as teachers input the raw scores
- Allow teachers to receive, view, and respond to direct inquiries and messages from parents/guardians regarding student academic performance, attendance, or document updates.

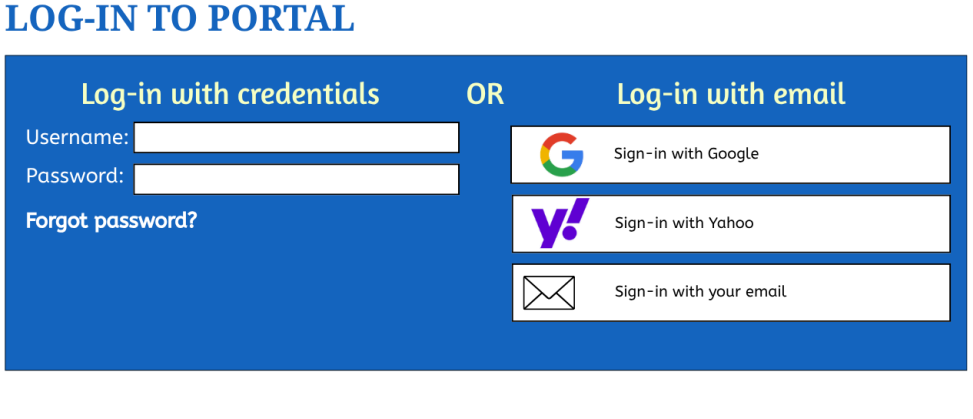
3. Admin and Document Features

- Allow admins to post and manage school-wide announcements and news updates on the dashboard.
- Provide role-based access controls allowing administrators to manage user accounts and assign specific subject loads and class sections to individual subject teachers.
- Format and export class records into standard CSV files that are compatible with LIS upload requirements.
- Automatically map student data and computed grades into official templates for official printing and archiving.
- Automatically generate a ticket and alert the admin/teacher dashboard whenever a new document is requested.
- Allow admins to update the status of a document request ticket as it moves through the process.
- Maintain a log that tracks any modifications made to finalized grades or document statuses for security and accountability purposes.

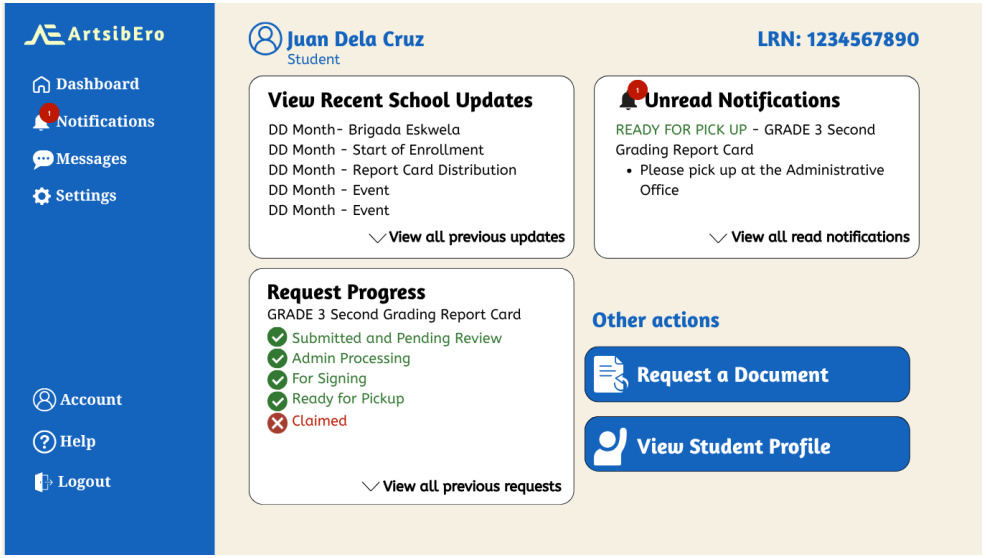
Preliminary UI Design

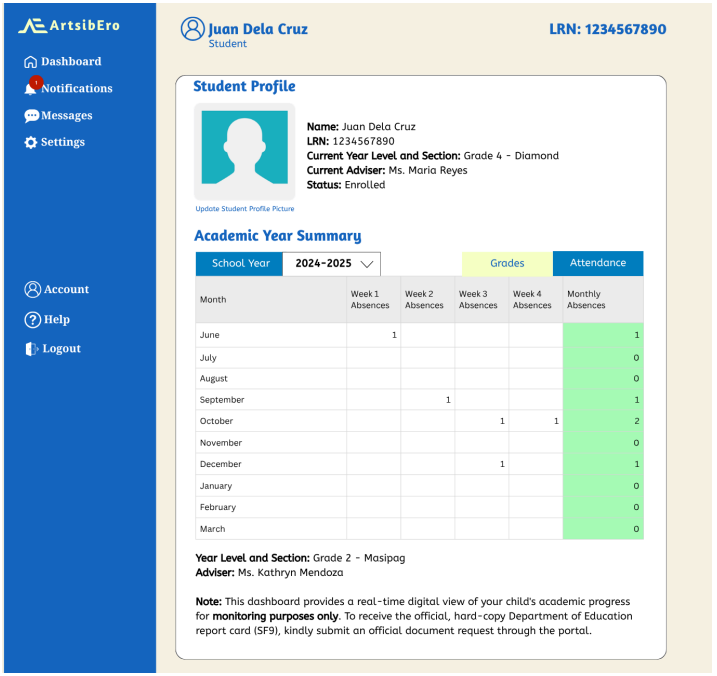
Announcements and log-in screen

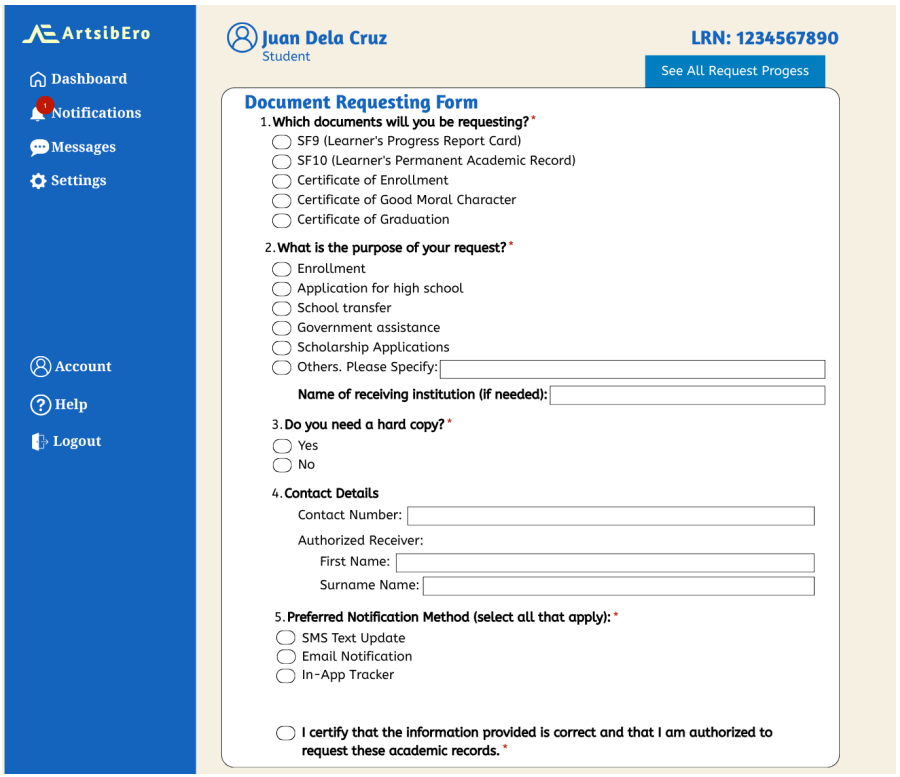


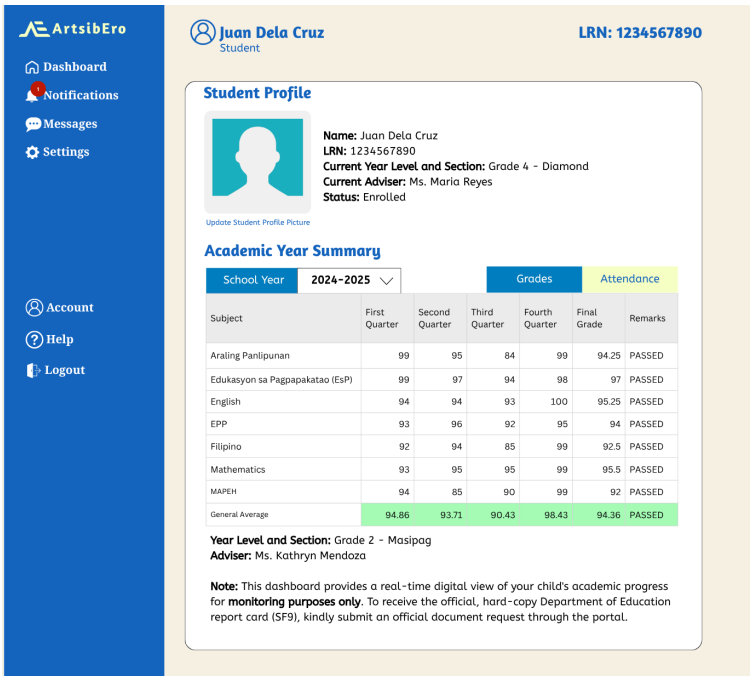


Parent Dashboard Tabs

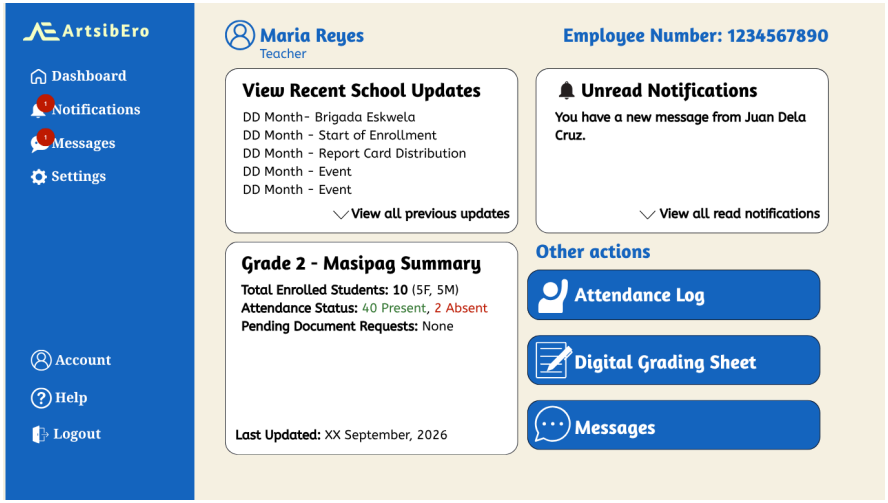


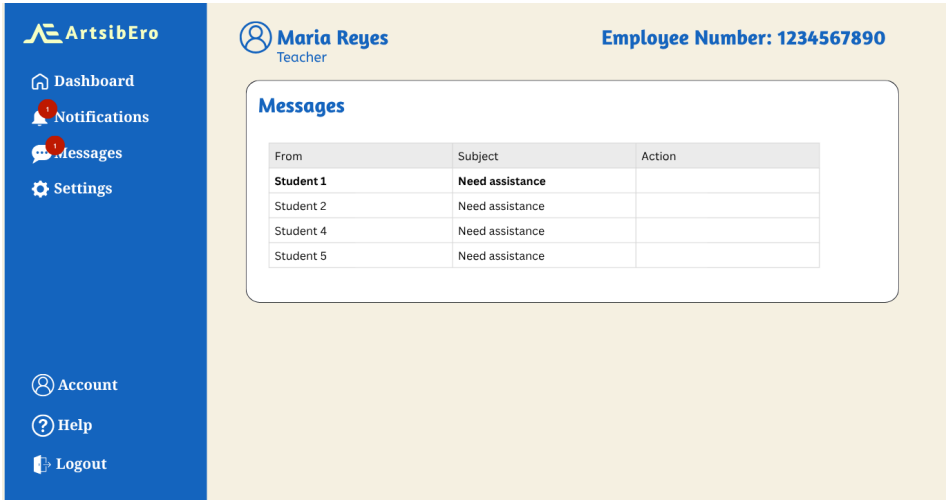


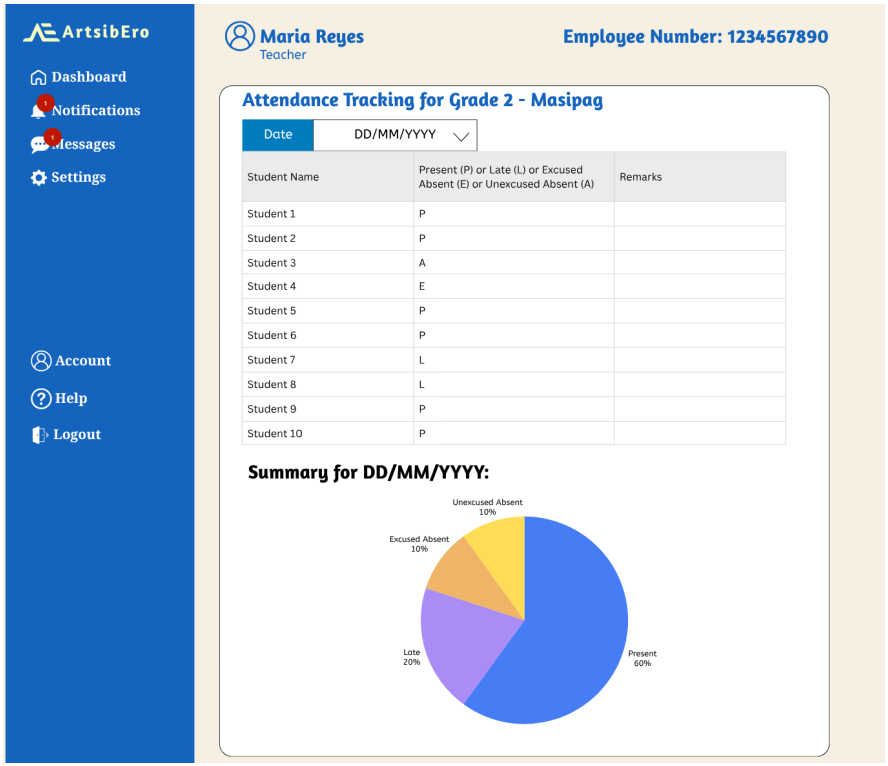


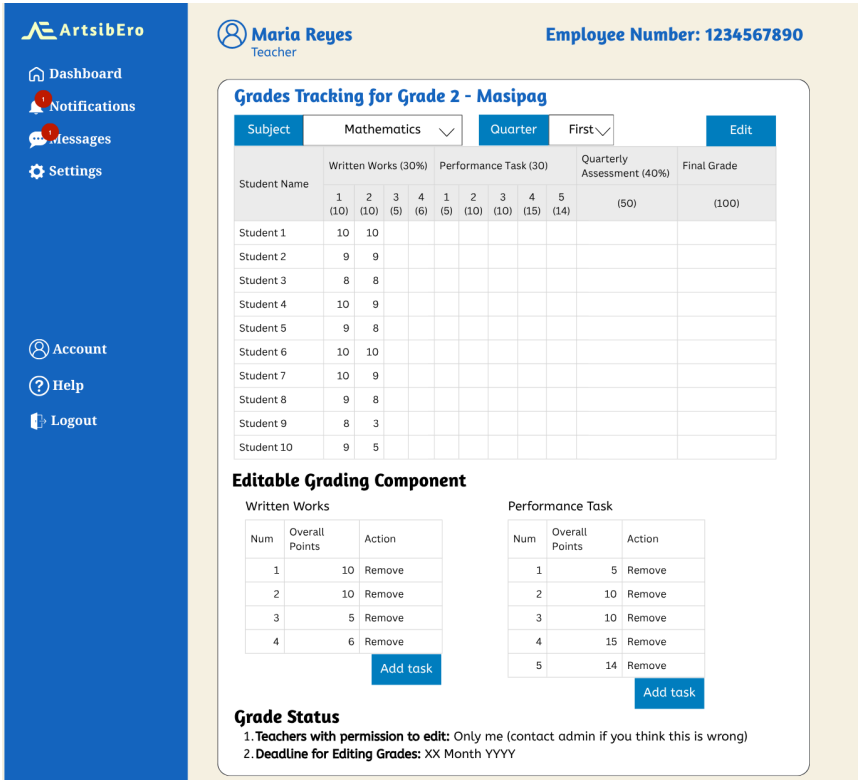


Teacher Dashboard and Tabs

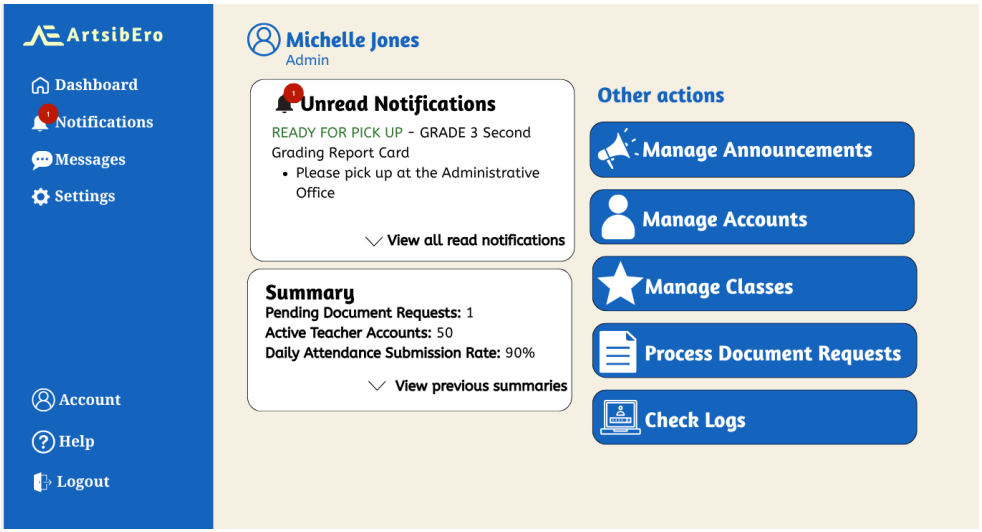


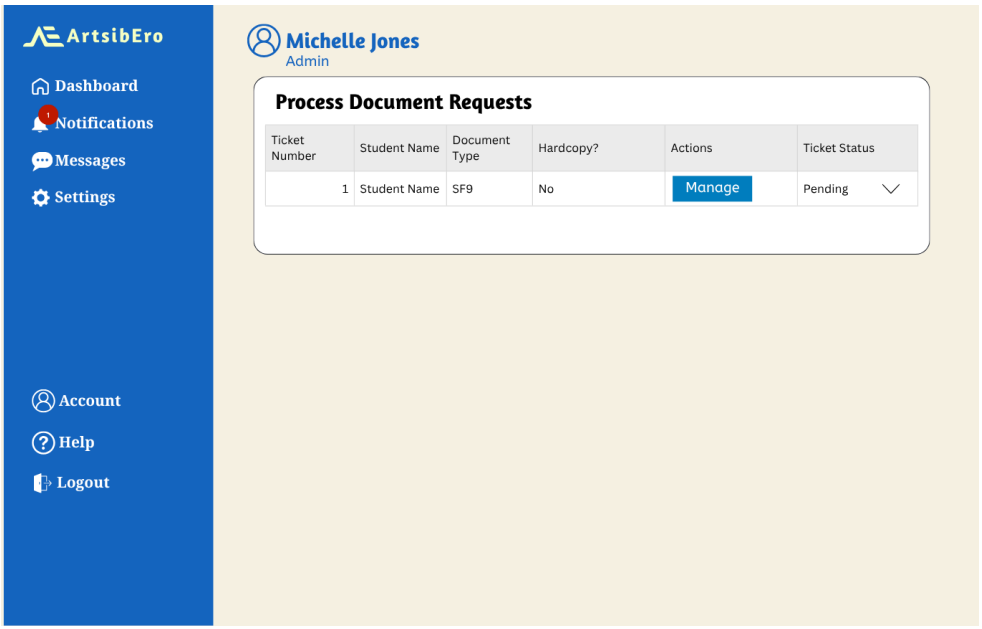


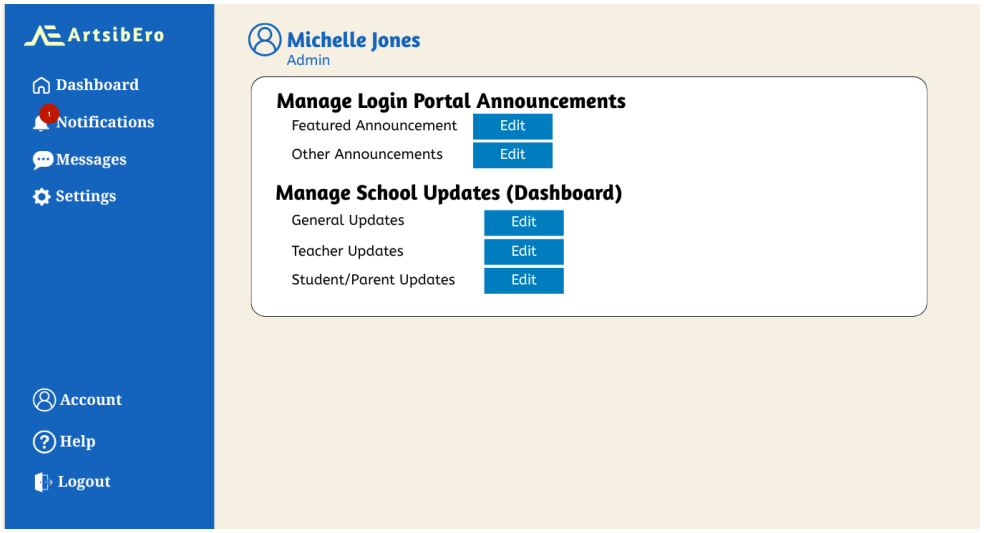


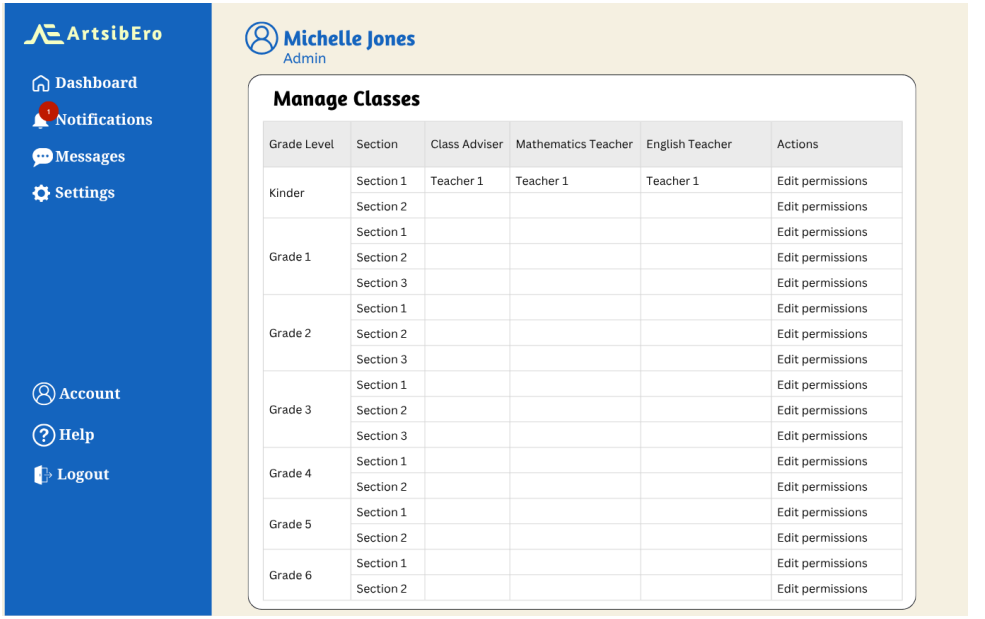


Admin Dashboard and Tabs









II. Plan of Work

Estimated Cost and Resources

The first heuristic to be used will utilize function points (FPs) as a metric of estimation as described by Pressman (2010). The FP values found in Table 1 for optimistic, likely, and pessimistic were derived from the expected user interactions with the website, and the chosen weights are the expected average relative complexities. The expected user interactions are as follows:

- External inputs: Authentication
- External outputs: Queried documents, Current progress, Received Messages
- External inquiries: Search inquiry, Progress inquiry, Template inquiry, Message inquiry
- Internal logical files: User preferences
- External interface files: Authentication, Relevant documents, Progress indicator

Information domain value	**Optimistic**	**Likely**	**Pessimistic**	**Estimate count**	**Weight**	**FP count**
External inputs	1	3	5	3	4	12
External outputs	3	4	5	4	5	20
External inquiries	4	6	8	6	4	24
Internal logical files	1	2	3	2	10	20
External interface files	3	4	5	4	7	28
Count total						104

Table 1. *Values used to calculate initial FP guess*

This initial FP guess is then paired with a value adjustment factor obtained by scaling the expected difficulty from various empirically determined factors that would affect the final complexity of the product. Our estimates can be seen in Table 2.

Factors	**Estimated value** (/5)
Backup and recovery	5
Data communications	3
Distributed processing	0
Performance critical	5
Existing operating environment	0
Online data entry	5
Input transaction over multiple screens	2
Master files updated online	5
Information domain values complex	2
Internal processing complex	1
Code designed for reuse	4
Conversion/installation in design	0
Multiple installations	0
Application designed for change	5
Value adjustment factor $(0.65 + 0.01 \times \Sigma F_i)$	**1.02**

Table 2. *Estimates for the various weighting factors*

We can now multiply the value adjustment factor to our initial FP guess, giving us an estimated 106.08 FPs. Since the average expected productivity is 6.5 FP/person-month (pm), the FP-estimated effort to complete this project is 16.32 pm. Note that pm, in this context, is a measure of the exerted effort of a person for one month.

To contrast with the last product-based heuristic, we also used a more process-based estimation, specifically, an object-oriented (OO) project estimation also as described by Pressman (2010). This estimation is used to build on the previous result to possibly give a much more accurate value meant to be more apt to the process to be used in development.

First, we consider the total number of use cases. In this case, we have the following possible queries: (1) sign up for the website, (2) log in to the website, (3) configure settings, (4) view alerts, (5) download document, (6) request document, (7) certify document, (8) check visual tracker, (9) edit document template, (10) manage permissions, and (11) sending messages. We can contextualize these using the various possible actors with their appropriate queries: guardian (1, 2, 3, 4, 5, 6, 8, 11), teacher (1, 2, 3, 4, 5, 7, 8, 11), or as an administrator (1, 2, 3, 4, 5, 7, 8, 9, 10, 11). This gives us a total count of 26 use cases which will be used to cross check the computed guess.

Second, we consider the model of the actual application. Figure 1 shows a preliminary relational model as it applies to this program.

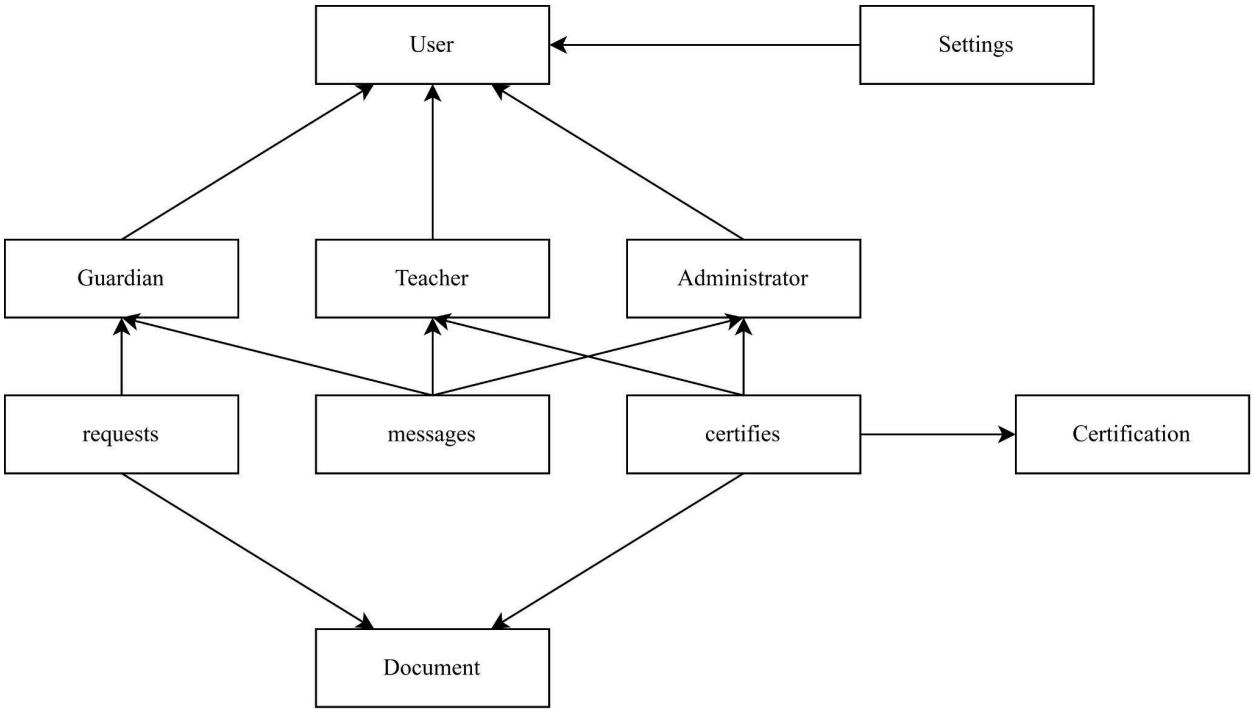


Figure 1. *Very basic relational model of the website*

The model can be used to estimate that we have seven (7) analysis classes. Since the interface is expected to be graphical but intuitive and easy to use, we use the prescribed GUI multiplier of 2.5 to estimate the number of expected support classes. This gives us an estimate of 24.5 total classes, which is in line with the number of use cases counted. Using an estimate of 24.75 classes/use-cases and an average expected productivity of 0.75 pm per class, the OO-estimated effort to complete this project is 18.56 pm, approximately two months more than the previous FP-based estimation.

As for the hardware estimations, since we plan to implement the system as a standard website, most of the hardware specifications will be gated by the specific modern web browser that the user plans on using. In general, a minimum of 800×600 screen resolution, 1 GB of RAM, an operating system capable of running the chosen web browser, and, an internet connection of at least 512 kbps, are needed (ICT Guide, 2026; Standard Technology Requirements; 2024). Since all interactions are based on the web, no storage space is needed to access the site but may be needed relative to the amount of documents the user may plan to download locally, as such, a preferred document viewer will be needed to open local documents.

Projected Schedule and Milestones

In general, we plan to utilize an evolutionary progression built around prototypes. Our overall milestones can summarized into the following:

1. Finalize project scope
2. Modeling of initial system architecture
3. Software reiterations
    a. Coding of prototype
    b. Internal testing and quality assurance
    c. Collection of stakeholder feedback
4. Full release of website application

As for the more specific feature-based milestones on the actual software construction, we have laid out our estimate for the thirteen major features as a Gantt chart in Table 3.

	Feature-based Tasks	***Week***															
		1		*2*		*3*		*4*		*5*		*6*		*7*		*8*	
A	Database and server setup																
B	Uploading and downloading queries																
C	Requesting and processing queries																
D	Templating queries																
E	Basic functional GUI																
F	Credentials authentication																
G	User onboarding																
H	Permission management																
I	Notification system																
J	Messaging system																
K	Document and Messaging security																
L	Basic accessibility settings																
M	GUI usability and accessibility pass																
N	GUI art pass																

Table 3. Gantt chart of initial schedule projections

Project Metrics

Various metrics (Pressman, 2010) will be used to track our project's progression. First, we will utilize the metrics used beforehand for the estimations as those will make for useful benchmarks. This includes the product-metrics, information domain values and FP, as well as, the OO-metrics, key and support classes. To support these, WebApp-based metrics will also be adapted to track the actual pages built and whether the internal logic and external inputs/queries have been bridged by the interface. Since the number of pages should be minimal, the metrics will focus more on the external systems interfaced, especially as they interact with static and dynamic content objects in the page.

Effort Breakdown

The following project table presents the planned tasks and functions derived from the project schedule, including their assigned team members and estimated effort. As assignments are not yet final, placeholder names were used to distribute assignments. Assuming a time spent of 8 weeks, with each person capable of spending a maximum of 5 days a week, we would have a maximum capacity of 120 person-days, with each person-day representing a standard workday, though not necessarily a full 8-hour block.

	Tasks and Modules	**Planned Start**	**Planned End**	**Assigned person(s)**	**Effort allocation (person-days)**
A	Database and server setup	**Wk 1**	**Wk 1**		6
	Database schema design			P1, P2	
	Server & environment setup			P3	
B	Uploading and downloading queries	**Wk 1**	**Wk 2**		12
	Document request forms			P1	
	Class record exports			P2	
	Announcements feed			P3	
C	Requesting and processing queries	**Wk 1**	**Wk 2**		12
	Request ticket			P1	
	Grade calculator			P2	
D	Templating queries	**Wk 2**	**Wk 2**		6

	Template mapping			P1, P2	
E	Basic functional GUI	**Wk 2**	**Wk 2**		6
	Parent view			P1	
	Employee/teacher view			P3	
F	Credentials authentication	**Wk 3**	**Wk 3**		6
	Parent login			P2	
	Employee/teacher login			P3	
G	User onboarding	**Wk 4**	**Wk 4**	All	4.5
H	Permission management	**Wk 3**	**Wk 4**		6
	Role definitions			P1	
	Role-based access			P3	
I	Notification system	**Wk 4**	**Wk 5**		4
	Ticket alerting			P2	
	Announcements notifications			P3	
J	Messaging System	**Wk 4**	**Wk 5**	P1	2
K	Document security	**Wk 4**	**Wk 5**		4
	Global modification log			P1	
L	Basic accessibility settings	**Wk 5**	**Wk 6**	P1, P3	4
M	GUI usability and accessibility pass	**Wk 6**	**Wk 7**	All	9
	UI consistency review			P3	
	Accessibility audit			P1, P2	
N	GUI art pass	**Wk 6**	**Wk 7**	All	4.5
O	Integration	**~Wk 6**	**Wk 7**	All	12
P	System Testing	**~Wk 7**	**Wk 8**	All	9
TOTAL					107
					89% of capacity

Rough estimations of the effort involved totals to 107 person-days, which is 89% of maximum capacity. This is notably lower than the effort estimated through the FP-estimated analysis (16.32 pm ≈ 326 pd) and OO-estimated analysis (18.56 pm ≈ 371 pd) previously. This difference reflects the smaller scope of our planned system compared to what a full production-grade EDMS would require, given the constraints of our timeline and team.

III. Project Risks

Some possible risks for the project are outlined below. Each risk is described according to their category, probability of occurrence, impact on project, and their risk mitigation, monitoring and management plan, all based on Pressman's (2010) risk management.

Risks may be categorized as **project size risk (PU), business impact (BU), client characteristics (CU), technology risk (TE), and staff size and experience (ST)**. Meanwhile, impact values may be **1-catastrophic, 2-critical, 3-marginal, 4-negligible**.

Risk Table

Risks	Category	Probability	Impact	RMMM
Requirements may change significantly in the middle of development	CU	High	1	Adopt iterative development workflow
Scope may shift based on client feedback	CU	High	2	Strictly confirm project scope with client early on; Focus on already agreed upon features
Project scope may be too wide for team size and constraints	PS	Moderate	2	Identify and breakdown essential features and components so others may be dropped
Unfamiliarity with possibly necessary tech stacks	TE/ST	Moderate	2	Start prototyping early to account for learning curve; Use familiar languages and tools as much as possible

Final system workflows may not fully align with actual user habits	CU	Moderate	3	Share prototypes and UI mockups with target users before finalizing designs; Analyze existing similar systems
Difficulty merging our system with client's existing records and workflows	CU/TE	Low	3	Design with existing processes in mind; Perform rigorous tests with similar data sets

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